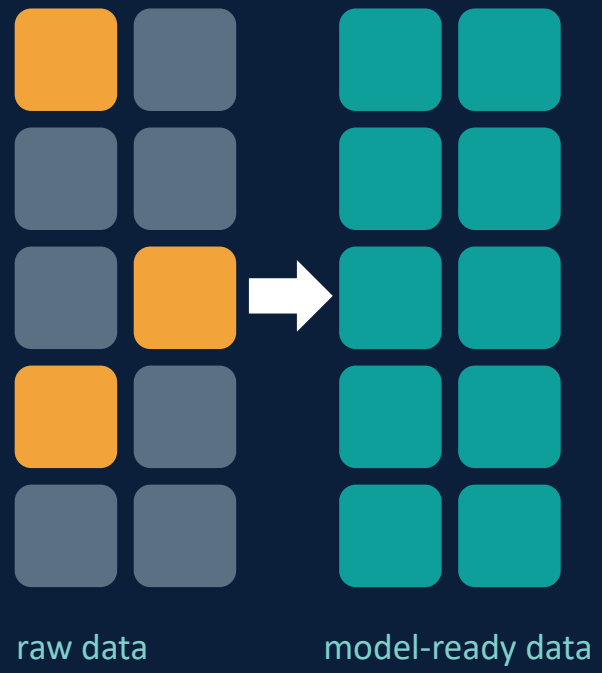
Machine Learning Foundations and Data Preparation

Practical Machine Learning | CDAC PGCP-AI

23 October 2026 | 4 hours theory + 3 hours lab



Today at a glance: 4 hours of theory, 3 hours of lab



Understanding checks

12+ short questions across the theory hours. Think first, answer on the next slide.

No model training today

Day 01 is about understanding ML and preparing data. Modelling starts in later sessions.

By the end of today you will be able to...

Explain

AI vs ML vs DL, supervised vs unsupervised, classification vs regression, and the words features, label, model.

Follow the workflow

Problem to data to preparation to training to evaluation to prediction, with train / validation / test sets.

Diagnose data

Spot missing, duplicate, inconsistent, invalid and suspicious values using the five quality dimensions.

Prepare data

Clean, encode, scale and validate a dataset, and explain every decision you made.

Opening problem: who will leave the bank next month?

Age	Tenure (yrs)	Transactions	Complaints	Will leave?
52	11	3	2	?
29	2	41	0	?
38	6	12	1	?
61	20	2	3	?

Illustrative customers



What would you look at?



Can you write rules for 5 million customers?



What if the data is wrong?

Why ML? Fixed rules cannot solve some problems

Spam filtering

Rules fail because spammers keep changing words, links and layouts.

Learns from: Past emails labelled spam / not spam

Card fraud detection

Fraud hides in combinations: amount, time, place, merchant and history.

Learns from: Past transactions labelled fraud / genuine

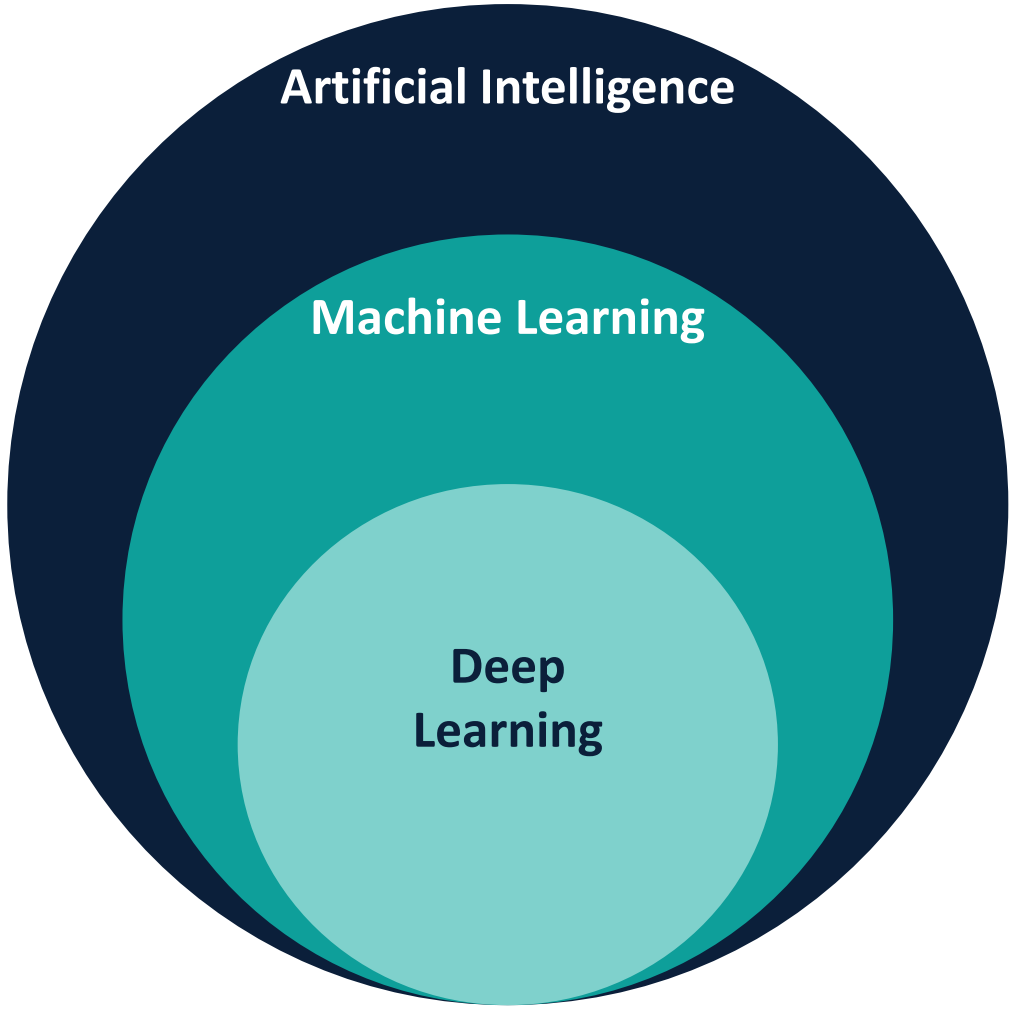
House price estimate

Price depends on area, locality, age, floor and market mood together.

Learns from: Past sales with their prices

Use ML when: rules are too many, rules keep changing, or the pattern is hidden in data and you have examples to learn from.

AI contains ML, and ML contains deep learning



Artificial Intelligence

Machines doing tasks that normally need human intelligence.

Example: Route planning, rule-based chess engine

Machine Learning

Systems that learn patterns from data instead of being told every rule.

Example: Spam filter, credit scoring

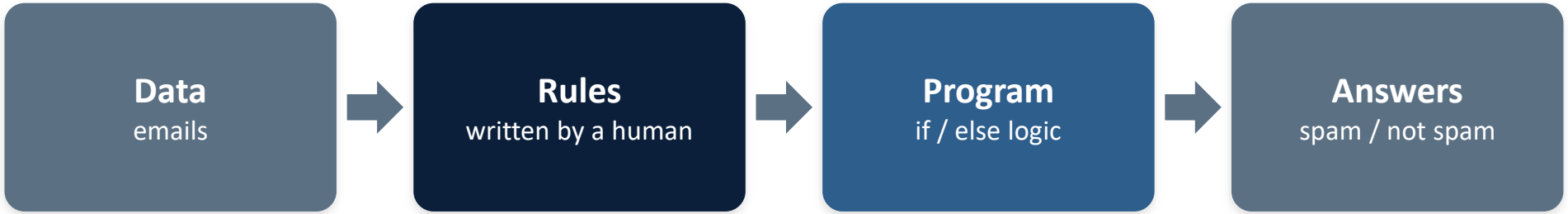
Deep Learning

ML using neural networks with many layers; strong on images, speech, text.

Example: Face unlock, speech to text

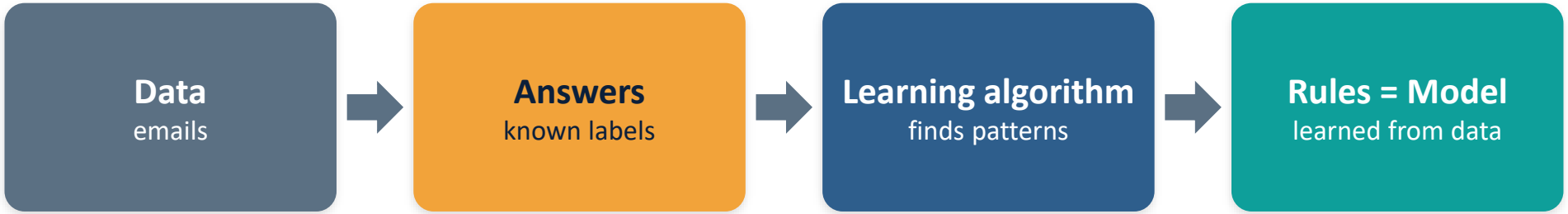
Traditional programming vs machine learning

Traditional programming



A programmer must think of every rule in advance.

Machine learning



The rules are learned from examples, then applied to new emails.

ML reverses the direction: examples in, rules out.

Supervised vs unsupervised learning

SUPERVISED

Learns from labelled examples

- Banking: predict loan default
- Retail: forecast next week's sales
- Healthcare: detect disease from scans
- Everyday: spam filter on your phone

UNSUPERVISED

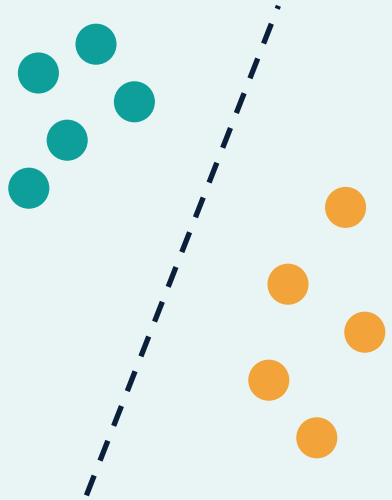
Finds structure without labels

- Banking: segment customers
- Retail: group similar shopping baskets
- Healthcare: find patient subgroups
- Everyday: cluster news stories

Classification: a category. Regression: a number

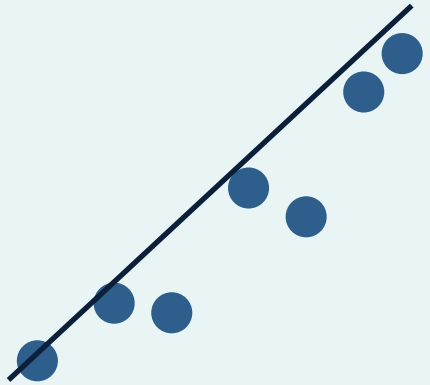
Classification: which class?

- Fraud or genuine?
- Will the customer churn: yes or no?
- Which disease category?



Regression: how much?

- House price in lakh rupees
- Sales next week
- Delivery time in minutes



Rule of thumb: look at the answer. A label means classification. A number means regression.

Features, label, record, algorithm and model

FEATURES (inputs) | LABEL / TARGET

one row = one record

Age	Income (Rs)	City	Account	Left bank?
34	5,40,000	Mumbai	Savings	**No**
52	9,10,000	Delhi	Premium	**No**
29	4,10,000	Pune	Current	**Yes**
41	7,20,000	Mumbai	Savings	**No**

Algorithm

The learning procedure (the recipe).

Model

What the algorithm produces after learning from your data (the dish).

Understanding Check 1: think first

Q1 | MCQ

A bank wants to label every card transaction as fraudulent or genuine.

Which problem type is this?

A) Regression
B) Classification
C) Clustering

Q2 | TRUE / FALSE

Customer segmentation, where no group labels exist, is supervised learning.

True or false? Justify your answer in one sentence.

Q3 | SCENARIO

Predicting house prices. Which are the features and which is the target?

Area (sq ft)	Bedrooms	Locality	Price (Rs lakh)
850	2	Andheri	145
1200	3	Baner	98

Answers: Check 1

A1 | B

Classification

The output is a category (fraud or genuine). Regression predicts a number; clustering has no labels at all.

A2 | FALSE

Unsupervised

No answer column exists. The algorithm discovers groups itself. Supervised learning needs labelled examples.

A3 | FEATURES

Features: area, bedrooms, locality
Target: price

Locality is text, so it must be encoded into numbers before many algorithms can use it (Hour 4).

Where machine learning is used today



Banking and finance

- Credit scoring (classification)
- Card fraud detection (classification)
- Customer segmentation (unsupervised)



Retail and e-commerce

- Demand forecasting (regression)
- Product recommendations
- Customer churn prediction (classification)



Healthcare

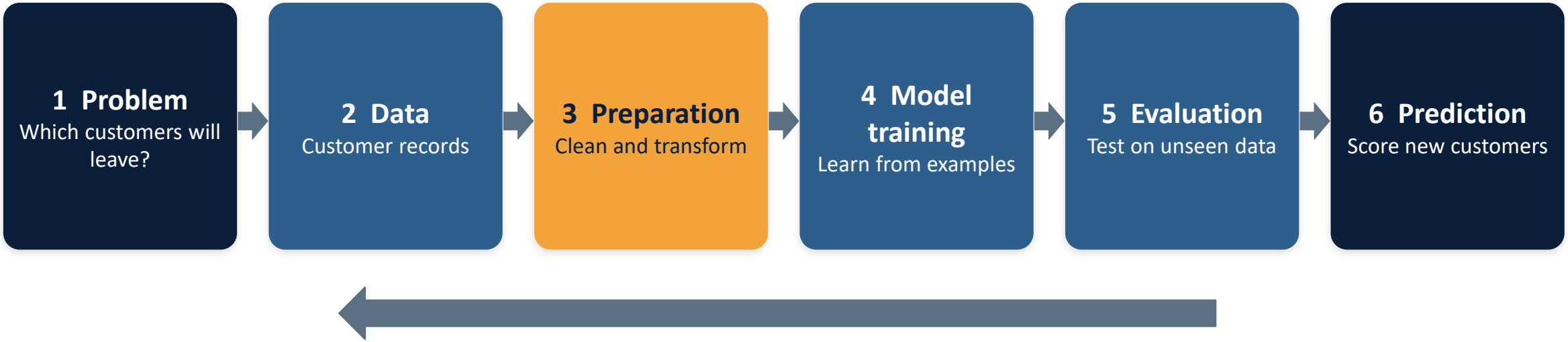
- Disease detection from scans
- Hospital readmission risk (classification)
- Appointment no-show prediction



Everyday technology

- Spam filtering in your inbox
- Maps: travel-time estimates (regression)
- Photo apps grouping faces (unsupervised)

The ML workflow: from problem to prediction



Poor result? Go back: better data, better preparation, different model

Today's lab focuses on step 3: Preparation

Inspect, clean, transform, validate, export. A model can only be as good as the data it learns from.

Garbage in, garbage out: data comes before the model

COLLECT



Core banking



Mobile app



Branch forms



Third-party data



PREPARE

- Remove duplicate records
- Fix missing values
- Standardise spelling and formats
- Convert text to numbers and dates
- Check impossible values
- Encode and scale



MODEL-READY

- Complete or flagged
- Consistent
- Valid ranges
- Unique records
- Numeric inputs

Garbage in, garbage out: a model copies the patterns in its data, including the mistakes.

Train, validation and test: three datasets, three jobs



Example of a typical split for 1,000 records: 700 / 150 / 150 (ratios vary)

Training set

The model learns patterns here.

Analogy: Practice questions

Validation set

Compare settings and models; tune.

Analogy: Mock exam

Test set

One final, honest score. Keep it untouched until the end.

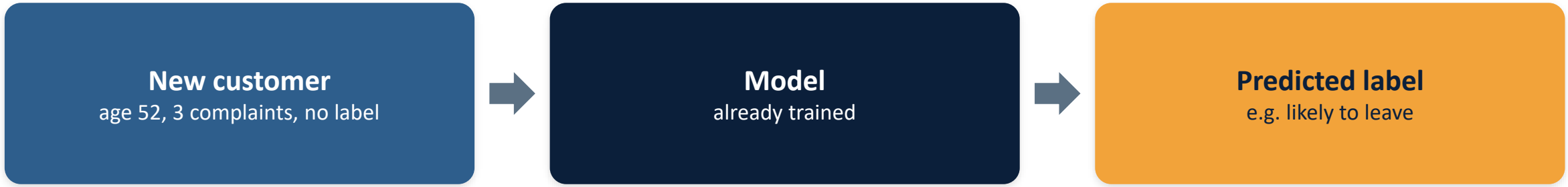
Analogy: Real exam

Training teaches the model; prediction uses it

PHASE 1: TRAINING



PHASE 2: PREDICTION

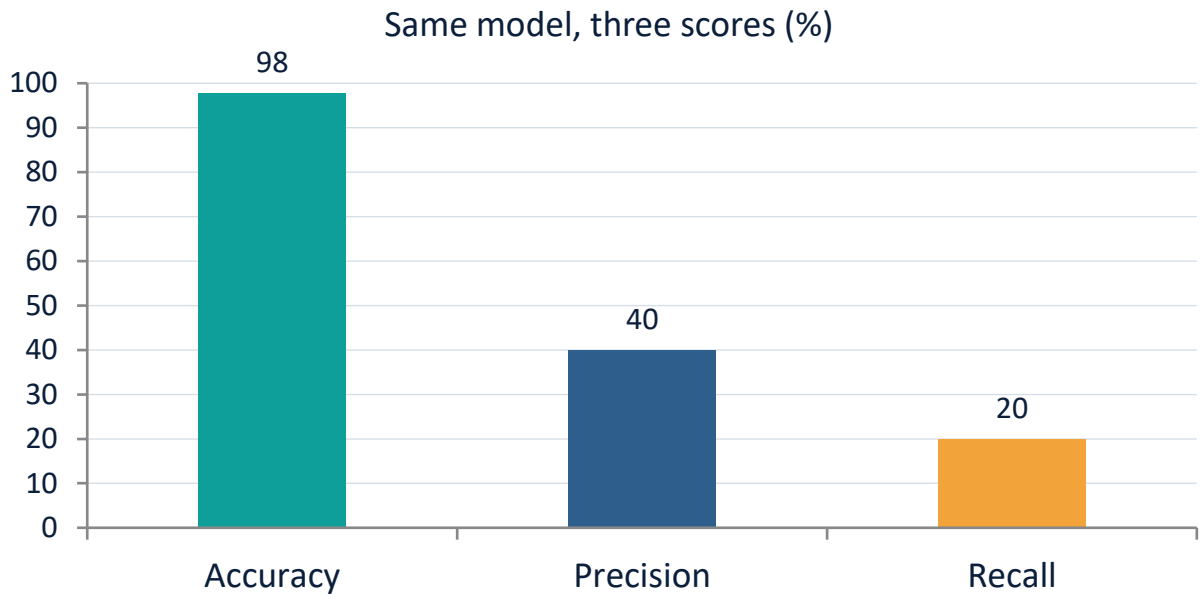


Illustrative example (no real model is built today).

Evaluating a model: accuracy is not the whole story

	Model says: FRAUD	Model says: GENUINE
Really FRAUD	**4** caught fraud (TP)	**16** missed fraud (FN)
Really GENUINE	**6** false alarm (FP)	**974** correctly cleared (TN)

1,000 transactions, 20 are fraud (illustrative)



Accuracy

correct / all = 978 of 1,000

Precision

of 10 flagged, 4 are fraud

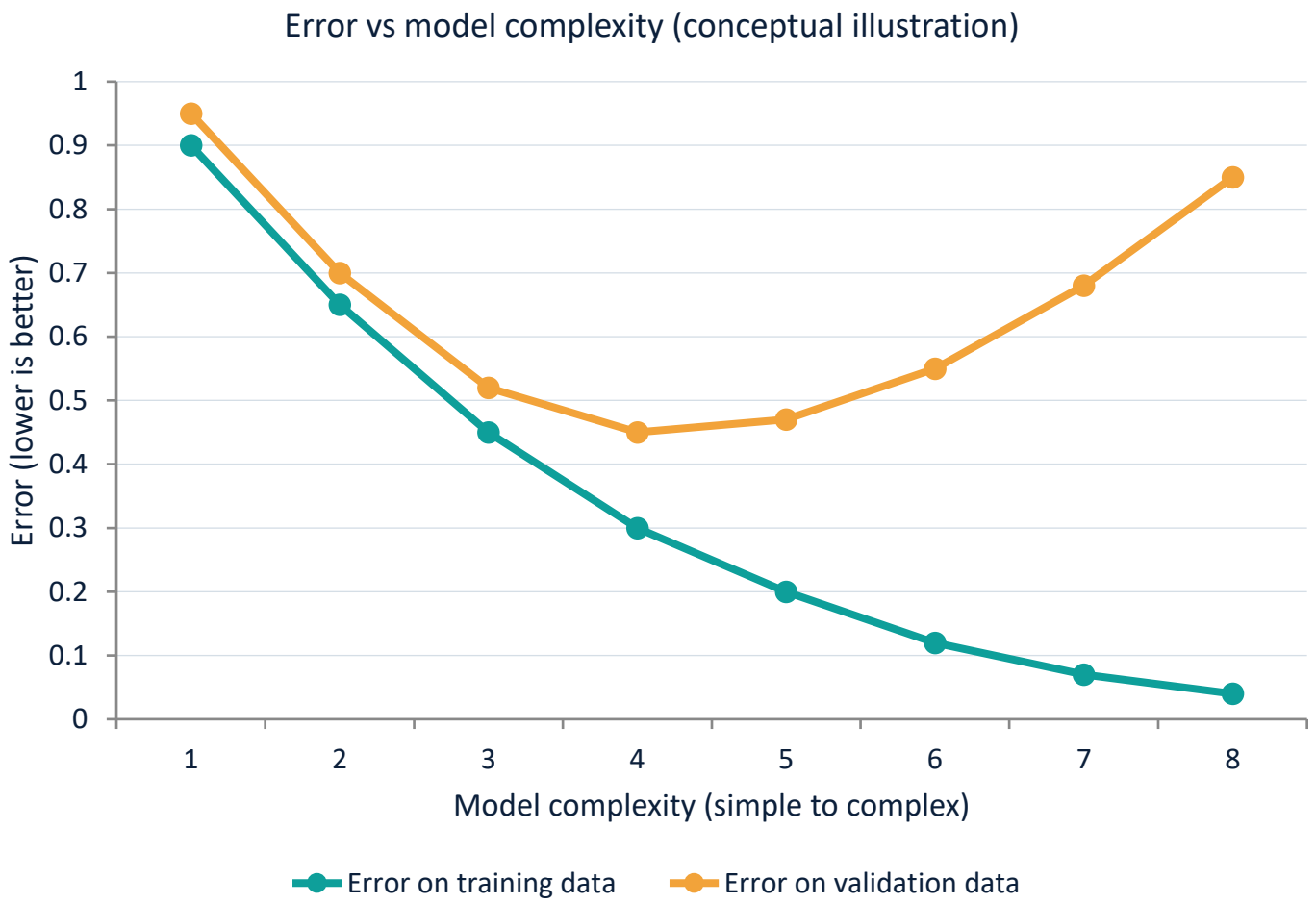
Recall

of 20 frauds, 4 caught

Prediction error

regression: predicted minus actual

Overfitting and underfitting (conceptual view)



Underfitting

Too simple. Poor on training AND new data.

Good fit

Balanced. Validation error is lowest.

Overfitting

Memorises noise. Great on training, poor on new data.

Understanding Check 2: workflow and testing

Q4 | SCENARIO

A model scores 99% on its training data but only 62% on unseen data.

What could be happening, and how could you confirm it?

Q5 | SCENARIO

A churn model is trained on records with wrong ages, duplicate customers and missing incomes.

Why is poor-quality input data a problem?

Q6 | TRUE / FALSE

'The test set is labelled, so we can use it again and again while tuning the model.'

True or false? Explain why test data must stay independent.

Answers: Check 2

A4 | OVERFITTING

The model memorised the training data, noise included.

Confirm: compare training vs validation scores. Reduce: more data, simpler model, regularisation.

A5 | GIGO

Garbage in, garbage out.

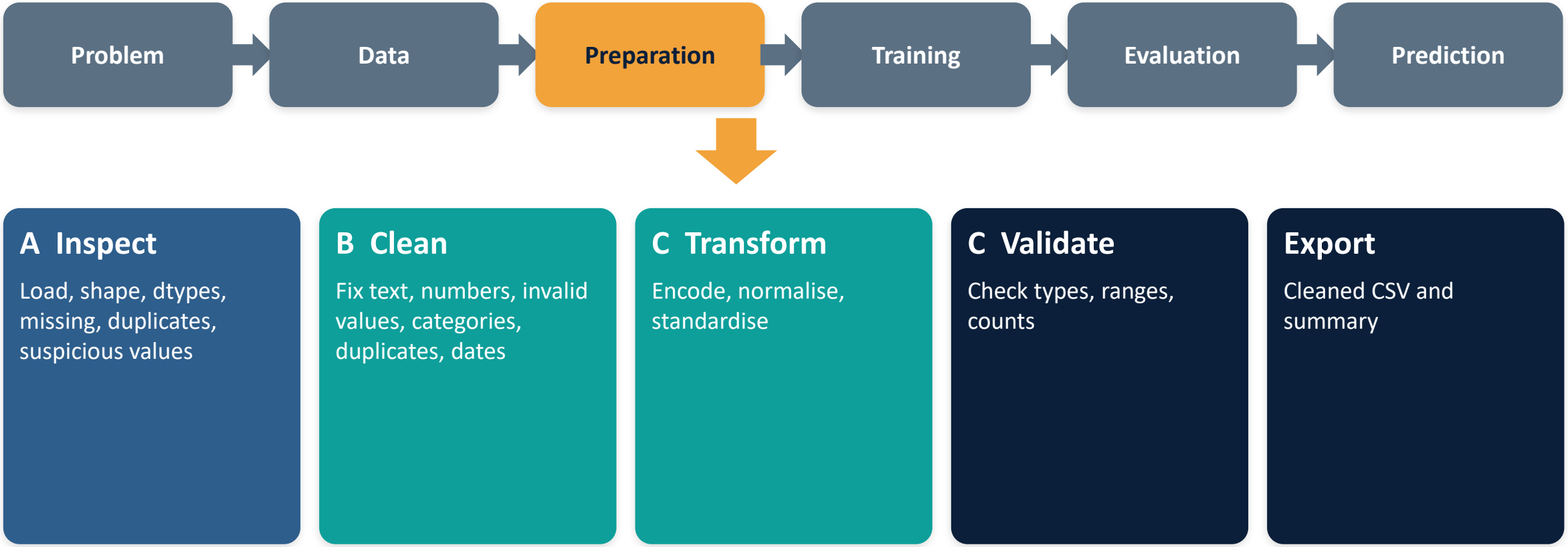
Wrong ages teach wrong patterns. Duplicates over-weight some customers. Missing values break many algorithms and bias results.

A6 | FALSE

Keep the test set independent.

It simulates unseen customers. If we tune on it repeatedly it leaks into our choices, and the final score becomes too optimistic.

Zoom in on Preparation: what happens inside step 3

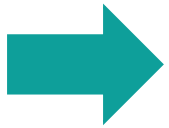


Next: Hours 3 and 4 teach the ideas. Then the lab turns them into practice.

Preprocessing: from raw to model-ready data

BEFORE (raw)

Age	City
29	pune
(missing)	Bombay
41	MUMBAI
250	Pune



AFTER (prepared)

Age	City
29	Pune
35	Mumbai
41	Mumbai
35	Pune

Algorithms need complete, numeric input

Bad data makes a bad model

Consistency lets us compare like with like

Structured and unstructured data

Structured

- Rows and columns, fixed schema
- Customer master, transactions, sensor logs
- Stored in databases, Excel, CSV
- Today's lab dataset

Unstructured

- No fixed schema
- E-mails, call recordings, scanned documents, photos
- Must be converted into numbers first
- Usually needs specialised methods (later sessions)

Our lab: a structured CSV of fictional bank customers (no real personal data).

Four kinds of columns

Numerical

Measured quantities. Arithmetic is meaningful.

Examples: age, annual income, transactions in 30 days

Typical tools: mean, median, scaling

Categorical (nominal)

Labels with no order.

Examples: city, account type, preferred channel

Typical tools: mode, counts, one-hot

Ordinal

Labels with a real order.

Examples: income band: Low < Medium < High

Typical tools: order-preserving codes

Date / time

Points in time.

Examples: signup date

Typical tools: year, month, tenure

Five dimensions of data quality



Completeness

Are values present?

EXAMPLE

missing age



Accuracy

Does it match reality?

EXAMPLE

age 250



Consistency

Same thing, same spelling?

EXAMPLE

Mumbai vs Bombay



Validity

Follows rules and formats?

EXAMPLE

negative transactions



Uniqueness

No unwanted repeats?

EXAMPLE

duplicate rows

Missing values and duplicate records

MISSING VALUES

ID	Age	Income
C101	34	540000
C102		620000
C103	29	
C104	41	810000

- Count them: how many, which columns?
- Ask why: not recorded, not applicable, lost?
- Decide: fill, flag or drop (Hour 4)

DUPLICATES

ID	Age	Income	Type
C102	38	620000	Savings
C102	**38**	**620000**	**Savings**
C207	45	410000	Current
C207	45	**950000**	Current

Exact duplicate

All columns identical. Drop the extras.

Conflict

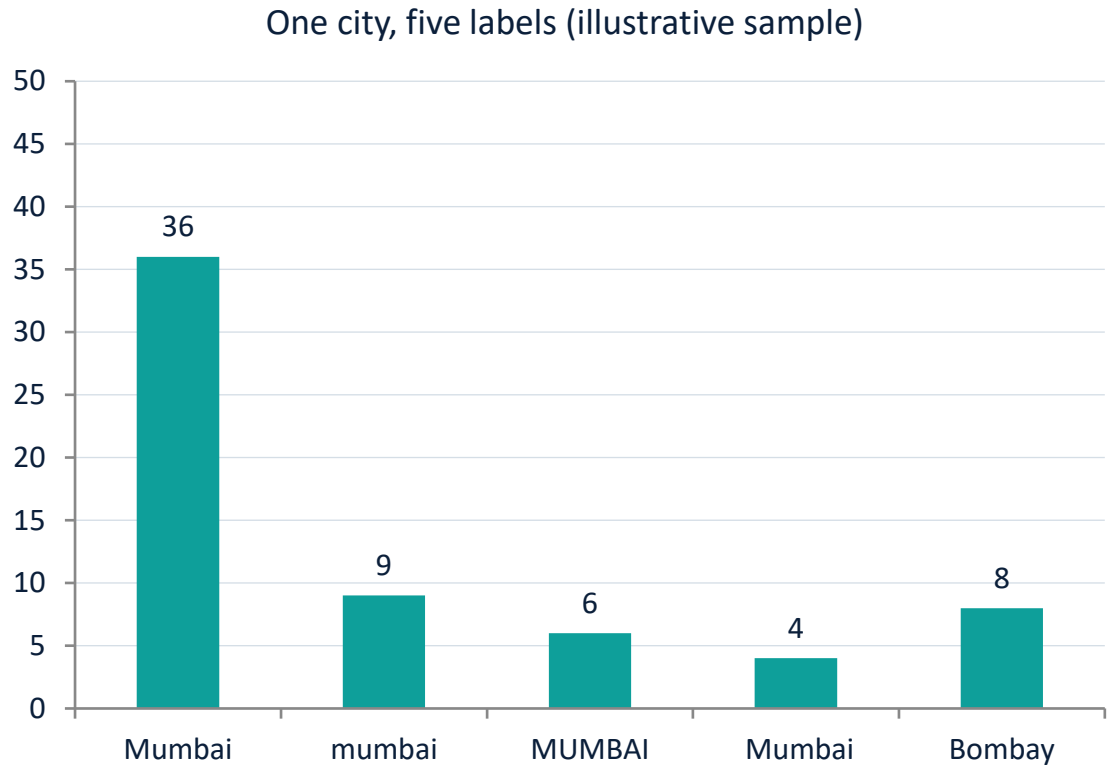
Same ID, different values. Flag, never delete both.

Wrong data types and inconsistent categories

NUMBERS STORED AS TEXT

Raw income (text)	Meaning
₹5,40,000	540000
INR 410000	410000
8,10,000	810000
34 yrs (age)	34

- Text cannot be averaged or scaled
- Remove symbols, convert, then check the dtype
- Keep the minus sign of negative numbers!



Counting 'Mumbai' gives 36, but the true total is 63.

Outliers: investigate before you change

Ages: 24, 32, 39, 41, 48 ... and 250



IQR fences: 15 to 65 => 250 is flagged

Statistics says 'unusual'. The business rule (age <= 100) says 'impossible'.

Identify first. Investigate the cause. Only then decide.

Invalid

- Breaks a hard rule: age 250, negative transaction count
- Cannot be true
- Set to missing, keep a flag

Suspicious

- Possible but unusual: income above Rs 1 crore
- Might be a VIP or a typo
- Flag and verify; do not change silently

EDA: look before you change anything

```
df.shape
```
rows and columns

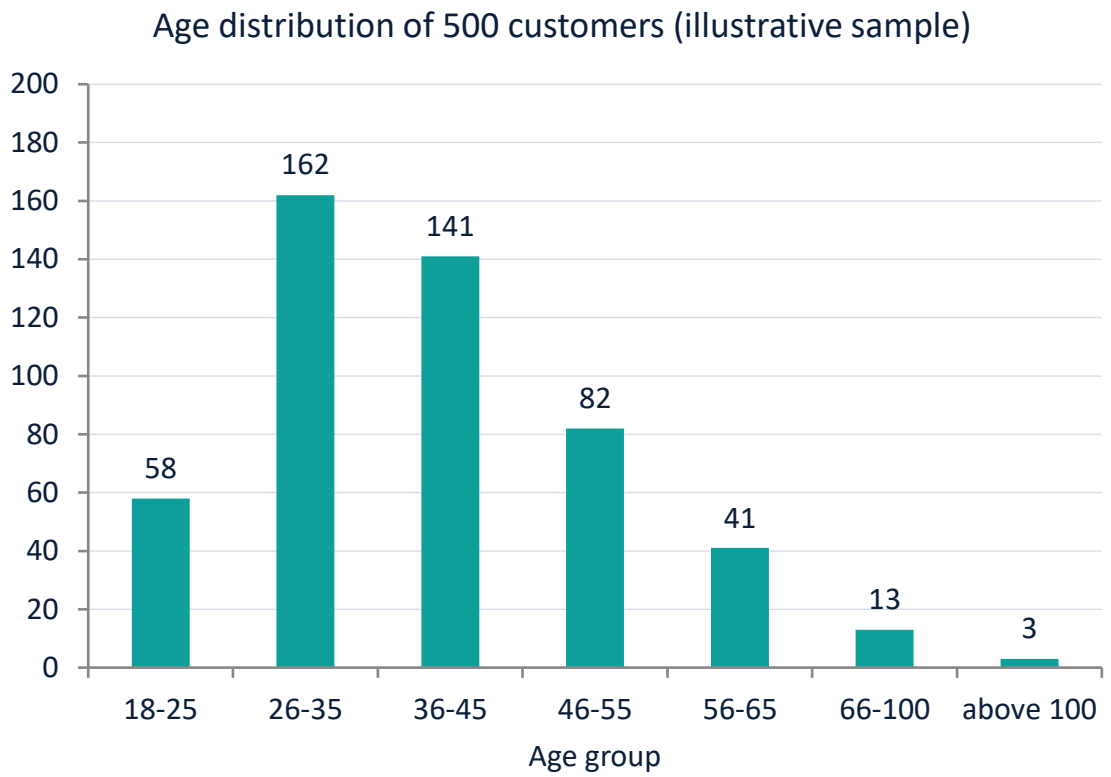
```
df.info()
```
types and non-null counts

```
df.describe()
```
min, max, mean, quartiles

```
df.isnull().sum()
```
missing values

```
df.duplicated().sum()
```
exact duplicates

```
df[col].value_counts()
```
category spellings



Three impossible ages are invisible in an average but obvious in a chart.

Understanding Check 3: find the problems

Q7 | INTERPRET THE TABLE

List every data-quality problem you can find. Which quality dimension does each one break?

Cust ID	Age	Income	City	Account	Txns (30d)
C101	34	₹5,40,000	Mumbai	Savings	12
C102		620000	mumbai	savings	9
C103	29	INR 4,10,000	Bombay	Current	-3
C102		620000	mumbai	savings	9
C105	250	750000	Pune	Gold	15
C106	41	8,10,000	Delhi	Premium	480

Answers: Check 3 (Q7)

Problem found	Where	Dimension
Missing value	Age of C102	Completeness
Exact duplicate row	C102 appears twice	Uniqueness
Different spellings / aliases / capitalisation	mumbai, Bombay, savings	Consistency
Number stored as text	Income with symbols, commas, INR	Validity (type / format)
Impossible age	250	Validity and Accuracy
Negative count	Txns = -3	Validity
Unexpected category	Account = Gold	Validity
Possibly extreme value	Txns = 480	Accuracy (suspicious)

Understanding Check 4: outliers and duplicate IDs

Q8 | OUTLIER

A customer dataset contains ages:

24, 32, 39, 41, 250, 48

What would you investigate before changing the value 250?

Q9 | DUPLICATE

Two rows have the same Customer ID but different account balances.

Should both records be deleted? Explain.

Answers: Check 4

A8 | INVESTIGATE

- Is 250 possible? Rule: age is 18 to 100, so it is invalid.
- Cause: typo (25 or 52), default value, source-system error?
- Related fields: date of birth, signup date.
- Then: flag, set to missing, impute. Never silently delete or cap.
- Impact: mean is 72.3 with the value, 36.8 without; median is 40.

A9 | NO

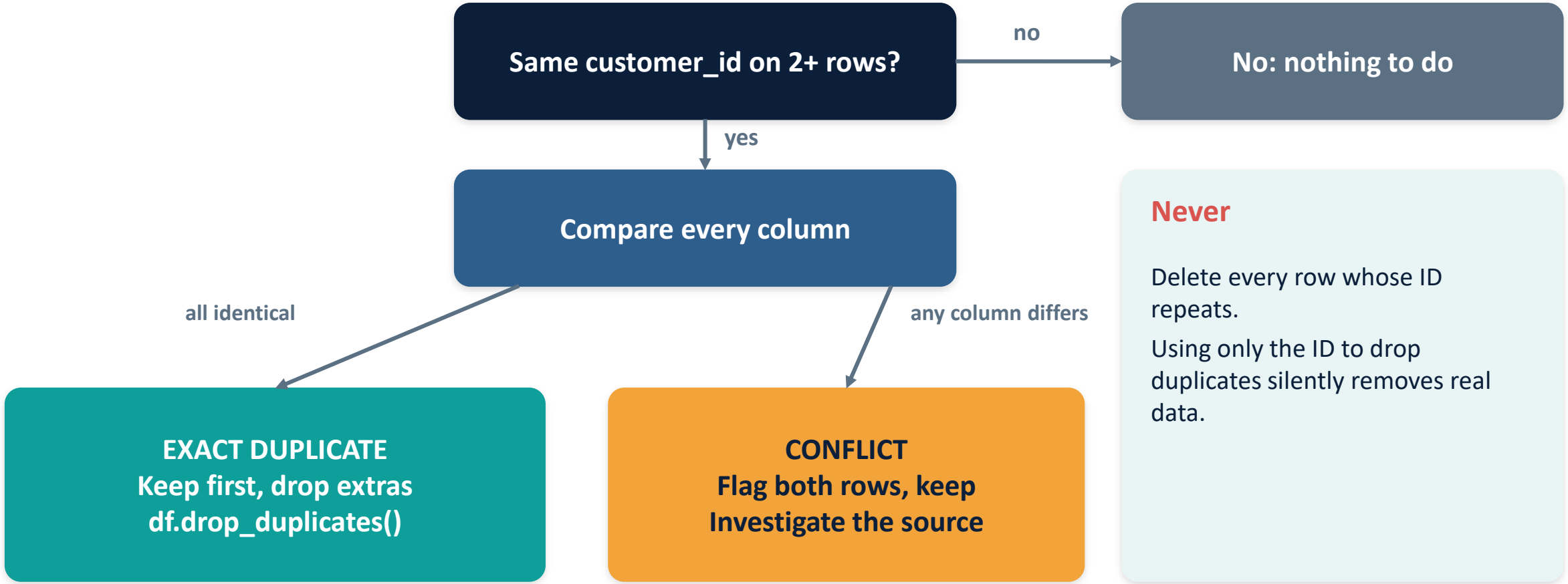
- **Do not delete both: that loses a real customer.**
- Investigate: timestamps, source system, which row is latest or verified.
- Until resolved: flag both rows and keep them.
- Exact duplicates (every column identical) are different: extra copies can be dropped.

Cleaning is a sequence: the order matters

- **B1** Work on a copy
- **B2** Fix text: spaces, case, aliases
- **B3** Convert numbers stored as text
- **B4** Flag invalid values, set to missing
- **B5** Missing / unexpected categories
- **B6** Remove exact duplicates
- **B7** Flag conflicting IDs
- **B8** Impute missing numbers
- **B9** Parse and validate dates

Raw data never changes. Flag before you fix. Explain every decision.

Duplicates: exact copies versus conflicting IDs

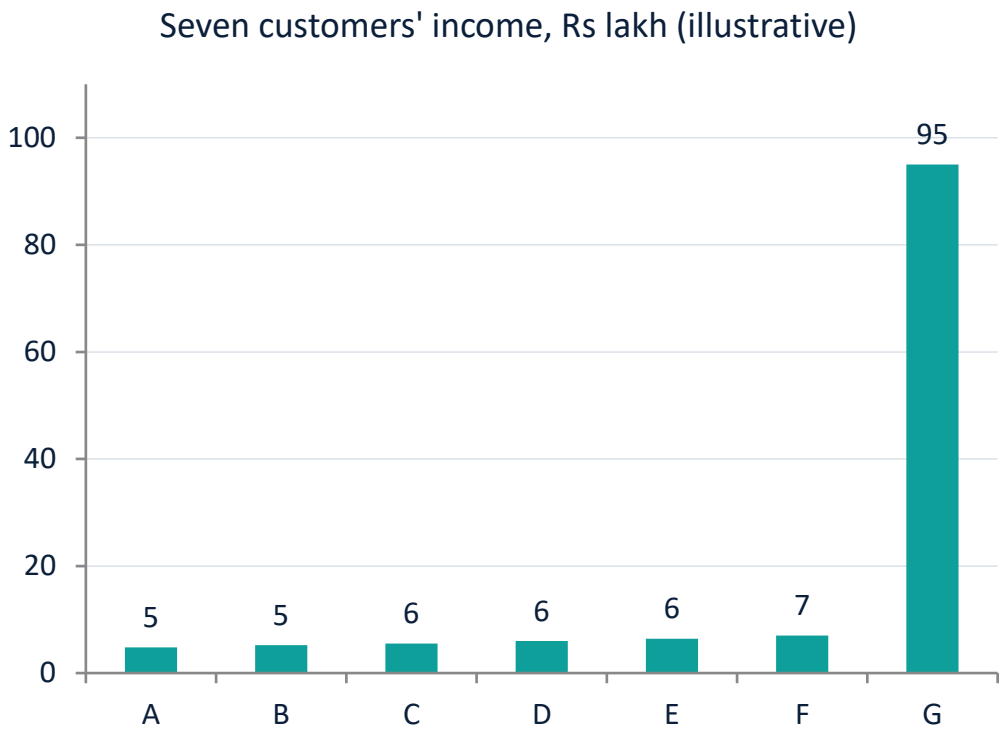


Never

Delete every row whose ID repeats.

Using only the ID to drop duplicates silently removes real data.

Missing numbers: mean, median or mode?



Mean = 18.6 | Median = 6.0

Fill AFTER duplicates and invalid values are handled.

Method	Good when	Drawback
Mean	Symmetric data, no extremes	Pulled by extreme values
Median	Skewed data or outliers	Ignores relations between columns
Mode	Categorical or discrete	Favours the majority; hides missingness

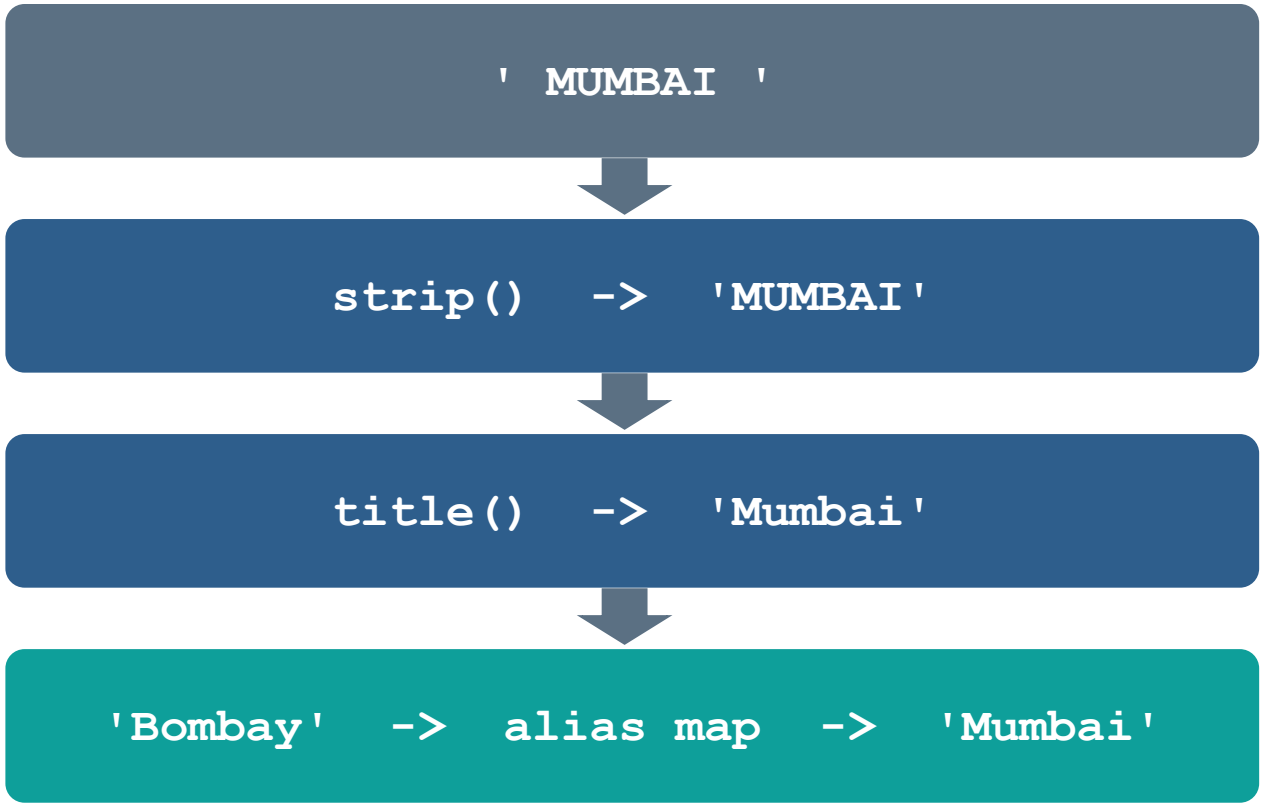
Before you fill

- How much is missing? 15 of 500 is 3%.
- Why is it missing? Random or systematic?
- Document the assumption you made.

Categories: fill gaps and make spelling consistent

Missing category: your options

- 'Unknown' placeholder: keeps the fact it was missing (our choice)
- Mode (most frequent): invents data, hides the gap
- Drop rows: loses other information in the row
- Unexpected values (e.g. 'Gold'): flag, then treat as Unknown




```
df["city"].str.strip().str.title().replace(aliases)
```

Convert numbers stored as text, and parse dates

Raw text	Number
₹5,40,000	540000
INR 410000	410000
-450000	-450000
34 yrs	34

```
text.str.replace(r"[^0-9.\-]", "", regex=True)
pd.to_numeric(..., errors="coerce")
```

Keep the minus sign, or -450000 silently becomes +450000.

Raw date text	Parsed date
2023-04-17	17 Apr 2023
17/04/2023 (DD/MM/YYYY)	17 Apr 2023
17-Apr-2023	17 Apr 2023
31/04/2024	**NaT (invalid) + flag**

```
pd.to_datetime(s, format="%d/%m/%Y",
               errors="coerce")
```

Try each known format, check the NaT count, then range-check.

Encoding: one-hot or ordinal

ONE-HOT (nominal)

city	Delhi	Mumbai	Pune
Pune	0	0	1
Mumbai	0	1	0
Delhi	1	0	0
Pune	0	0	1

Use when

- One-hot: no natural order (city, channel)
- Ordinal: real order (Low < Medium < High)

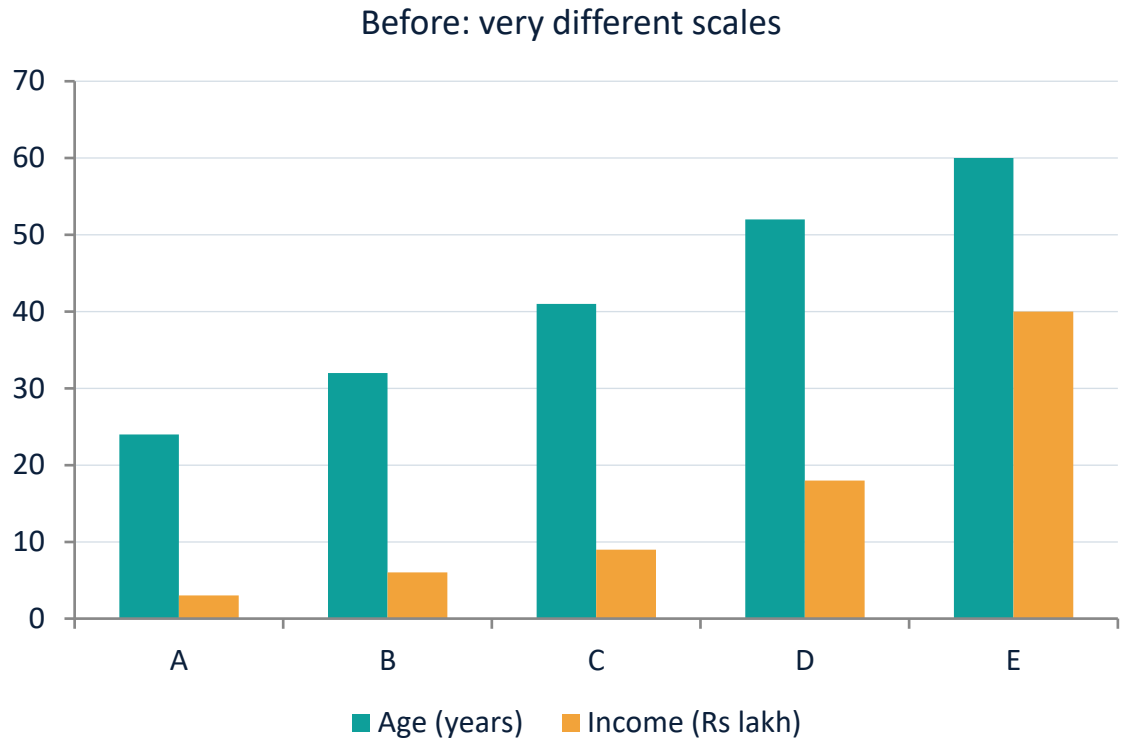
ORDINAL (ordered)

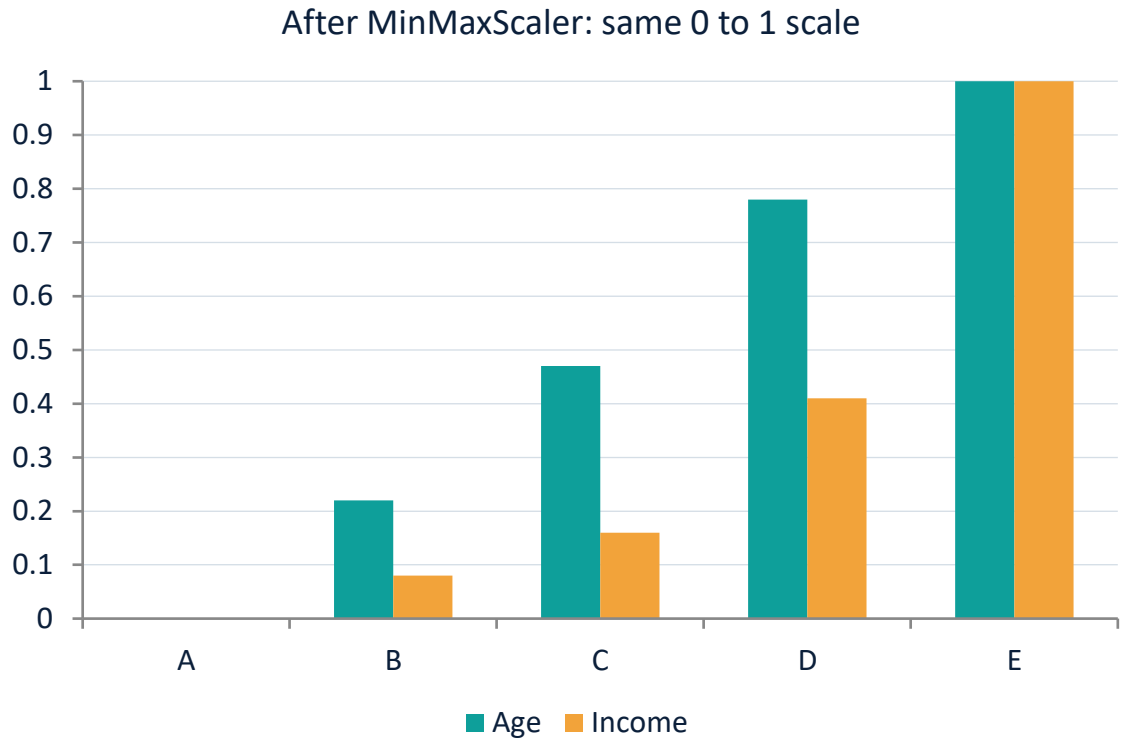
income band	code
Low	0
Medium	1
High	2

Watch out

- Never number nominal labels: Pune is not 'greater' than Delhi
- Clean first: MUMBAI and Mumbai would become two columns
- One-hot adds a column per category

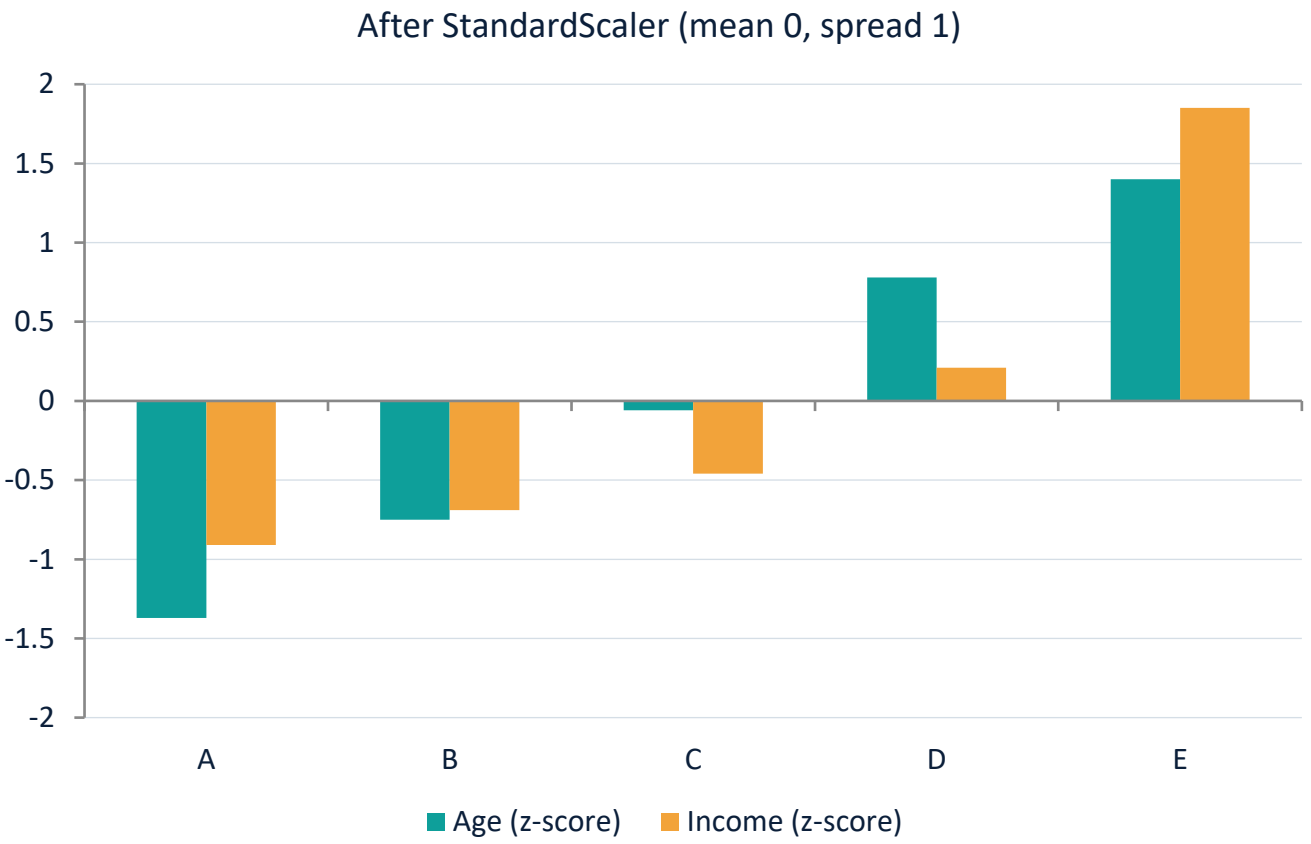
Normalisation with MinMaxScaler: squeeze into 0 to 1





new value = (x - minimum) / (maximum - minimum). Smallest becomes 0, largest becomes 1. Illustrative customers, not the lab data.

Standardisation with StandardScaler: centre and rescale



Same five illustrative customers as the previous slide.

z = (x - mean) / std

How many standard deviations from the average?

Good to know

- Mean 0, standard deviation 1
- Not limited to a fixed range
- Extreme values still stand out as large +/- scores

Normalisation vs standardisation: how to choose

	Normalisation (MinMaxScaler)	Standardisation (StandardScaler)
Idea	Rescale to a fixed range, usually 0 to 1	Centre on 0, divide by spread
Formula in words	(x - min) / (max - min)	(x - mean) / standard deviation
Result	Bounded: 0 to 1	Mean 0, std 1; not bounded
Good when	A bounded range is needed; no extreme values	Data roughly bell-shaped; extremes present
Assumption	Min and max are representative	Mean and spread are representative
Drawback	One extreme value squeezes all others	Extremes still distort mean and std

Neither scaler fixes bad data. Investigate outliers first; fit scalers on training data only.

Outliers: identify, investigate, then decide

1 Identify
statistics and business rules

→

2 Investigate
typo? unit? genuine?

→

3 Decide
document the reason

Error and true value known
Correct it

Impossible value
Set to missing and flag, then impute

Possible but unusual
Keep and flag; verify with the business

Never
Delete silently

In the lab: suspicious values are kept and flagged, so you can see their effect on scaling.

Validation: prove the data is ready

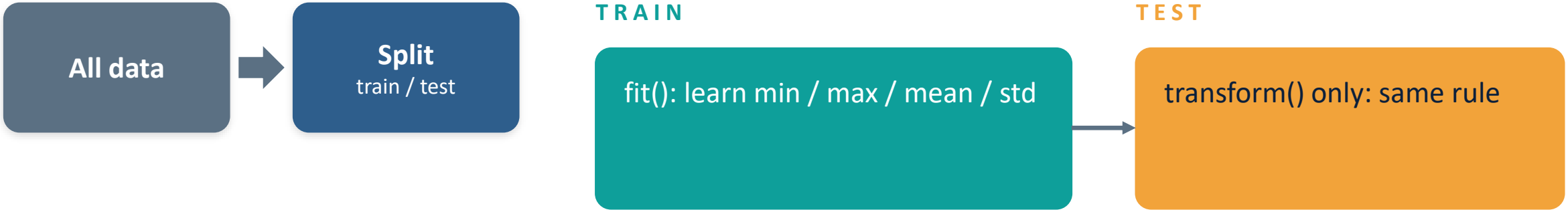
- Required columns present
- Numeric columns are numeric (dtype)
- Values inside agreed ranges (e.g. age 18 to 100)
- No exact duplicates left
- Row-count change fully explained
- Remaining missing / suspicious values reported

```
assert clean["age"].between(18, 100).all()
assert not clean.duplicated().any()
```

Mini demo dataset	Raw	Cleaned
Rows	6	5
Missing cells	2	0
Exact duplicates	1	0
Invalid ages	1	0
Rows carrying a flag	0	1

Illustrative 6-row demo, not the lab dataset. Flags that remain are reported, not hidden.

Data leakage: learn rules from training data only




```
scaler.fit(X_train)                    # learn from TRAIN only
X_train_s = scaler.transform(X_train)
X_test_s  = scaler.transform(X_test)   # same rule, no re-
learning
```

Leakage

fit on test = test data shapes the rule

Applies to everything that learns from data: imputation values, scalers, encoders. No model is trained today; this is the rule you will apply from Day 02.

Understanding Check 5: choose the technique

Q10 | CHOOSE THE METHOD

A dataset contains 500 customer records. Fifteen records have missing age values.

Which method would you use to handle these values, and what would influence your choice?

Q11 | CHOOSE THE ENCODING

Customer data has a city column: Mumbai, Delhi, Pune, Chennai...

Which encoding technique would you choose, and why?

A) Ordinal numbers 0, 1, 2, ... B) One-hot C) Drop the column

Answers: Check 5

A10 | MEDIAN (usually)

- 15 of 500 is only 3%: filling is reasonable.
- Median: robust to outliers and skew. Mean only if symmetric.
- Dropping rows loses information and can bias if not random.
- Choice depends on: % missing, distribution, outliers, why missing.
- Document it; consider a missing-flag column.

A11 | B ONE-HOT

- City is nominal: no natural order.
- Ordinal 0, 1, 2 would suggest Pune is greater than Mumbai.
- Clean first (Bombay to Mumbai, casing, spaces).
- Drawback: many cities means many columns.

Understanding Check 6: scaling and leakage

Q12 | SCENARIO

Two features: age (18 to 75) and annual income (Rs 1,00,000 to Rs 40,00,000).

What problems could occur if a distance-based model uses them without scaling?

Q13 | PREDICT

A MinMaxScaler is fitted on the ages [20, 30, 40, 60].

What scaled value does it give to the age 50?

Q14 | FIND THE MISTAKE

What is wrong with this code?

```
scaler = StandardScaler()
X_train_s =
scaler.fit_transform(X_train)
X_test_s  =
scaler.fit_transform(X_test)
```

Answers: Check 6

A12 | INCOME DOMINATES

- Distances are driven by the feature with the largest numbers: income.
- Customer A (30, Rs 5,00,000) vs B (60, Rs 5,10,000): distance is about 10,000.04. The 30-year age gap is almost invisible.
- Scale both features to comparable ranges.

A13 | 0.75

(x - min) / (max - min)
= (50 - 20) / (60 - 20)
= 30 / 40 = 0.75

Min is 20 and max is 60 because those are the fitted ages.

A14 | LEAKAGE

- fit_transform(X_test) re-learns the rule from test data.
- Test data leaks into preprocessing and train and test use different rules.
- **Fix: scaler.transform(X_test)**

Lab: Data Preparation, Cleaning and Transformation

Plan: 180 min

- Stage A Explore (60)
- Stage B Clean (60)
- Stage C Transform and validate (60)
- Each hour: explain, demo, you try, review

Setup checklist

- Open Jupyter or Colab
- Open 04_Day01_Student_Exercise.ipynb
- Keep 06_customer_data_dirty.csv beside it
- Keep 07_Data_Dictionary.xlsx open for the business rules
- Run the first code cell

The dataset

- 9 columns: ID, age, income, city, account type, channel, signup date, transactions, average amount
- Fictional customers, planted problems
- Rules: copy first, flag before delete, explain decisions

Demo: load and inspect a table (6-row mini example)

OBJECTIVE **See rows, columns and data types of a small table**

cust	age	income	city
D1	29	4,50,000	'Pune'
D2	35	₹620000	'pune '
D3	NaN	510000	'Mumbai'
D4	41	NaN	'Mumbai '
D2	35	₹620000	'pune '
D6	250	720000	'Delhi'

```
demo.shape
demo.head(3)
demo.dtypes
```

OUTPUT

```
(6, 4)

  cust   age   income    city
0   D1  29.0  4,50,000    Pune
1   D2  35.0   ₹620000   pune
2   D3   NaN    510000  Mumbai

cust   object | age   float64
income  object | city  object
```

Interpretation

Income is stored as text (symbol and commas). Age is float because a value is missing. Newer pandas prints 'str' instead of 'object'.

YOUR TURN **Exercises A1 to A5 on the real dataset (load, head, shape, info, describe)**

Demo: count missing values and find duplicates

OBJECTIVE **Count missing values; separate exact duplicates from repeated IDs**

```
demo.isnull().sum()

demo.duplicated().sum()

demo[demo["cust"].duplicated(keep=False)]
```

OUTPUT

```
cust     0
age      1
income   1
city     0

1

  cust   age   income   city
1   D2  35.0  ₹620000  pune
4   D2  35.0  ₹620000  pune
```

Interpretation

One exact duplicate: D2 appears twice, identical in every column. If the incomes differed it would be a conflict: flag it, never drop it.

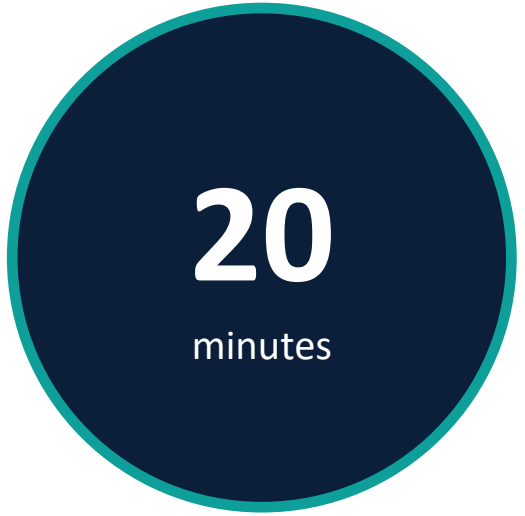
YOUR TURN **Exercises A6 and A7 on the real dataset: missing percentage per column, exact duplicates, repeated IDs**

Your turn: explore the real dataset (A1 to A7)

EXERCISE CHALLENGE

Identify all columns containing missing values and calculate the missing percentage of each column.

A1	Load the CSV into df
A2	Display the first 10 records
A3	Find the number of rows and columns
A4	Identify the data type of each column
A5	Generate descriptive statistics
A6	Count missing values and percentages
A7	Exact duplicates and repeated customer IDs



While you work

- Notebook sections 1 to 4
- Use the Hint boxes
- Stage A: inspect only, change nothing
- Ask when you are stuck after trying